

30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

Statistics

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

Question 1

Find the mean for data 2 (f=3), 4 (f=2), 6 (f=5).

Answer:

Sum $f = 10$; sum $fx = 44$. Mean = sum fx /sum $f = 4.4$.

Question 2

Find the mean for data 5 (f=2), 10 (f=4), 15 (f=4).

Answer:

Sum $f = 10$; sum $fx = 110$. Mean = sum fx /sum $f = 11$.

Question 3

Find the mean for data 1 (f=4), 3 (f=3), 5 (f=3).

Answer:

Sum $f = 10$; sum $fx = 28$. Mean = sum fx /sum $f = 2.8$.

Question 4

Find the mean for data 8 (f=1), 12 (f=5), 16 (f=4).

Answer:

Sum $f = 10$; sum $fx = 132$. Mean = sum fx /sum $f = 13.2$.

Question 5

Find the mean for data 10 (f=3), 20 (f=3), 30 (f=4).

Answer:

Sum $f = 10$; sum $fx = 210$. Mean = sum fx /sum $f = 21$.

Question 6

Find the mean for data 4 (f=5), 6 (f=2), 8 (f=3).

Answer:

Sum $f = 10$; sum $fx = 56$. Mean = sum fx /sum $f = 5.6$.

Question 7

Find the mean for data 7 (f=2), 9 (f=6), 11 (f=2).

Answer:

Sum $f = 10$; sum $fx = 90$. Mean = sum fx /sum $f = 9$.

Question 8

Find the mean for data 12 (f=4), 14 (f=4), 16 (f=2).

Answer:

Sum $f = 10$; sum $fx = 136$. Mean = sum fx /sum $f = 13.6$.

Question 9

Find the mean for data 3 (f=1), 6 (f=4), 9 (f=5).

Answer:

Sum $f = 10$; sum $fx = 72$. Mean = sum fx /sum $f = 7.2$.

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Question 10

Find the mean for data 25 (f=2), 30 (f=5), 35 (f=3).

Answer:

Sum f = 10; sum fx = 305. Mean = sum fx/sum f = 30.5.

Question 11

For grouped data, l=10, h=10, f₁=12, f₀=8 and f₂=6. Find the mode.

Answer:

Mode = $l + \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)}h = 14$.

Question 12

For grouped data, l=20, h=10, f₁=15, f₀=9 and f₂=7. Find the mode.

Answer:

Mode = $l + \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)}h = 24.2857$.

Question 13

For grouped data, l=30, h=10, f₁=18, f₀=11 and f₂=10. Find the mode.

Answer:

Mode = $l + \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)}h = 34.6667$.

Question 14

For grouped data, l=40, h=10, f₁=20, f₀=12 and f₂=8. Find the mode.

Answer:

Mode = $l + \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)}h = 44$.

Question 15

For grouped data, l=50, h=10, f₁=14, f₀=7 and f₂=9. Find the mode.

Answer:

Mode = $l + \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)}h = 55.8333$.

Question 16

For grouped data, l=0, h=5, f₁=16, f₀=8 and f₂=10. Find the mode.

Answer:

Mode = $l + \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)}h = 2.85714$.

Question 17

For grouped data, l=15, h=5, f₁=22, f₀=14 and f₂=11. Find the mode.

Answer:

Mode = $l + \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)}h = 17.1053$.

Question 18

For grouped data, l=25, h=5, f₁=19, f₀=10 and f₂=12. Find the mode.

Answer:

Mode = $l + \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)}h = 27.8125$.

Question 19

For grouped data, l=35, h=5, f₁=13, f₀=6 and f₂=7. Find the mode.

Answer:

Mode = $l + \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)}h = 37.6923$.

Statistics - Continued

Question 20

For grouped data, $l=45$, $h=5$, $f_1=17$, $f_0=9$ and $f_2=8$. Find the mode.

Answer:

Mode = $l + [(f_1 - f_0) / (2f_1 - f_0 - f_2)]h = 47.3529$.

Question 21

State the main formula, rule or theorem used in Statistics.

Answer:

Mean = $\sum fx / \sum f$; grouped-data median and mode use their stated formulas.

Question 22

What do the important symbols or quantities in Statistics represent?

Answer:

f is frequency; class mark is the midpoint; modal class has the greatest frequency.

Question 23

Write the correct method for solving a Statistics numerical.

Answer:

Make a frequency table, compute required columns carefully, identify the correct class, then use the formula.

Question 24

Which unit, sign convention or condition must be checked in a Statistics answer?

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

Question 25

Why must the final answer in Statistics be checked against the given condition?

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

Question 26

What is one common error in Statistics questions?

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

Question 27

How should full working be shown in a Statistics numerical?

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

Question 28

When is a diagram, table or graph useful in Statistics?

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

Statistics - Continued

Question 29

How can a student decide which formula to use in Statistics?

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

Question 30

What makes a high-scoring answer in Statistics?

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: [CBSE Mathematics curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); [NCERT textbooks](#).

ONE SHOT MASTERY